

Semantic-Guided Imitation Learning with History-Conditioned Feedback

Abstract—

Index Terms—Imitation learning, semantic representations, visuomotor control, residual feedback.

I. Introduction

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III. Problem Formulation and Theoretical Analysis

This section studies conditions for improving task performance from limited demonstrations. Starting from the relationship between imitation error and task success, we examine two paths for improvement: preserving information needed to distinguish the task consequences of different actions, and using new observations and execution history to produce effective corrections.

We first establish conditions under which action error bounds task-failure probability and explain why improved coverage in representation space alone does not guarantee better task performance. We then bound representation loss in terms of task-relation estimation error, providing a conditional basis for semantic guidance and relation supervision. Finally, we characterize the potential decision value of new observations and history, deriving a sufficient condition for feedback to improve task completion.

A. Visuomotor Imitation Learning

We study offline visuomotor imitation learning from a finite, fixed set of robot demonstrations, for manipulation tasks whose completion can be assessed from visual history. The policy generates control commands from visual and proprioceptive observations and available execution history. Training is based on this fixed demonstration dataset, with demonstrated actions providing the primary imitation target, whereas deployment performance is measured by the probability of task completion over a finite horizon.

1) *System and Observations:* Consider a robotic system governed by a single decision policy that performs a given manipulation task through the joint actions of its actuators. The joint robot-environment state $s_t \in \mathcal{S}$ evolves under the command $u_t \in \mathcal{A}$ and gives rise to observation o_t . The observation $o_t = (I_t^{1:V}, q_t)$ comprises multi-view images $I_t^{1:V} = (I_t^1, \dots, I_t^V)$ and proprioceptive information q_t , where $V \geq 1$ is the number of cameras and I_t^v is the image from the v th view. Over a finite horizon of T control steps, this interaction is represented by the following partially observed controlled process:

$$\begin{aligned} s_0 &\sim p_0, \\ s_{t+1} &\sim P(\cdot \mid s_t, u_t), \quad 0 \leq t < T, \\ o_t &\sim O(\cdot \mid s_t), \quad 0 \leq t \leq T. \end{aligned} \quad (1)$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A} \subseteq \mathbb{R}^{d_u}$ is the admissible command set specified by the chosen robot control interface, d_u is the command dimension, p_0 is the initial-state distribution, and P and O are the transition and observation kernels, respectively.

2) *Offline Demonstrations and Policy:* The policy learns from a fixed demonstration set $\mathcal{D} = \{\zeta_i\}_{i=1}^N$ for the same manipulation task. The i th demonstration trajectory ζ_i records observations and demonstrated control commands:

$$\zeta_i = (o_0^i, u_0^{E,i}, o_1^i, \dots, u_{T_i-1}^{E,i}, o_{T_i}^i). \quad (2)$$

Here, N is the finite number of demonstrations, $T_i \geq 1$ is the number of control steps in the i th trajectory, and the superscript E denotes demonstrated commands.

The information available to the deployed policy π is represented by the history $\mathcal{H}_t$. This history includes observations $o_{0:t}$ obtained up to time t , previously executed commands $u_{0:t-1}$, and other runtime records before the current decision. The policy defines a distribution over admissible control commands, with $u_t \sim \pi(\cdot \mid \mathcal{H}_t)$. The command history is empty at $t = 0$.

3) *Learning and Task Objectives:* For the following analysis, we consider deterministic policies acting on the available history, with π_E denoting the expert policy. Let ν denote the population distribution underlying the demonstrations, where $(t, h) \sim \nu$ specifies a control step and its available history. We quantify the imitation objective by the expected squared Euclidean deviation from the expert's action:

$$R_\nu(\pi) := \mathbb{E}_{(t,h) \sim \nu} \|\pi(h) - \pi_E(h)\|^2. \quad (3)$$

Training seeks to reduce this population risk through an empirical imitation loss on the finite demonstrations; the specific training losses are defined in Section IV.

The deployment objective is to complete the task within the finite horizon. The measurable task-completion function $\Phi : \mathcal{S}^{T+1} \rightarrow \{0, 1\}$ equals 1 for state trajectories satisfying the task's terminal or temporal requirements and 0 otherwise. This defines the completion indicator Ψ and the task-completion probability $J(\pi)$ of the deployed policy π :

$$\Psi = \Phi(s_{0:T}), \quad J(\pi) = \mathbb{P}_\pi(\Psi = 1). \quad (4)$$

Here, $\mathbb{P}_\pi$ is the trajectory probability law induced jointly by the system model and the complete deployed policy π . Learning aims to reduce $R_\nu(\pi)$, whereas deployment aims to increase $J(\pi)$. A lower imitation risk does not by itself guarantee a higher task-completion probability; the following analysis examines conditions relating these objectives.

4) *Visual Assessability*: Under a fixed camera configuration and evaluation horizon T , the evaluation domain $\Omega_{\text{eval}} \subseteq \mathcal{S}^{T+1} \times \mathcal{I}^{T+1}$ consists of prespecified pairs of physical trajectories and visual histories consistent with the system model and visibility conditions, where $\mathcal{I}$ is the space of multi-view image tuples $I_t^{1:V}$.

Assumption 1 (Visual assessability): There exists a measurable visual decision function $\mathcal{V} : \mathcal{I}^{T+1} \rightarrow \{0, 1\}$ such that, for every $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$,

$$\Phi(s_{0:T}) = \mathcal{V}(I_{0:T}^{1:V}). \quad (5)$$

Assumption 1 specifies visual identifiability of completion within the evaluation domain, without requiring full-state observability. The subsequent analysis considers policy executions and counterfactual action comparisons for which $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$ almost surely.

B. From Imitation Error to Task Performance

Whether action error bounds task failure depends on where deviations occur and their subsequent task consequences. We establish this connection and show that transferring action risk to deployment depends jointly on representation-level coverage and conditional action-error discrepancy.

1) *Task-Performance Difference Between Policies*: We compare a policy π with a reference policy π_0 through their task-failure probabilities and express the difference as a sum of expected single-step action consequences. Each consequence is defined by comparing alternative actions at the same history while following π_0 afterward [1], [2]. This decomposition provides the basis for relating imitation error to task performance.

We consider deterministic policies acting on the available history. Given a history $\mathcal{H}_t = h$ and a candidate action $a \in \mathcal{A}$, define the action-value function as the probability of task failure ($\Psi = 0$):

$$Q_t^{\pi_0}(h, a) := \mathbb{P}^{a, \pi_0}(\Psi = 0 \mid \mathcal{H}_t = h). \quad (6)$$

Here, $\mathbb{P}^{a, \pi_0}$ denotes the probability law obtained by taking action a at time t and following π_0 afterward.

When the current action is also selected by π_0 , this reduces to the value function of π_0 , $V_t^{\pi_0}(h) := Q_t^{\pi_0}(h, \pi_0(h))$. The effect of choosing a instead of $\pi_0(h)$ is therefore measured by

$$A_t^{\pi_0}(h, a) := Q_t^{\pi_0}(h, a) - V_t^{\pi_0}(h). \quad (7)$$

Thus, $A_t^{\pi_0}(h, a) > 0$ indicates an increase in failure probability, whereas $A_t^{\pi_0}(h, a) < 0$ indicates a decrease.

To relate these action-level consequences to differences in task performance between policies, we require the following common comparison conditions.

Assumption 2 (Common task and environment): The compared policies share the same initial distribution and task-completion criterion. Given a history h and an action a , the next available history follows a conditional distribution that does not otherwise depend on the policy. The reference policy π_0 is defined on all histories involved in the comparison.

Let $F(\pi) = 1 - J(\pi)$ denote the task-failure probability of policy π . To evaluate π relative to π_0 , we set $a = \pi(h)$ and

aggregate the resulting action consequences along trajectories generated by π .

Applying the finite-horizon performance-difference identity [1]–[3] to these one-step action effects gives

$$F(\pi) - F(\pi_0) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) \right]. \quad (8)$$

Here, $\mathbb{E}_\pi$ denotes expectation over trajectories generated by π . Equation (8) evaluates action effects at the situations actually visited by the evaluated policy π . Its proof is given in Appendix A-A. This relation allows task-performance differences to be bounded through the consequences of action deviations.

2) *When Action Error Bounds Failure Probability*: Taking the expert policy as the reference, we show that the imitation risk $R_\nu(\pi)$ bounds task-failure probability $F(\pi)$ when demonstrations sufficiently cover deployment histories and action deviations have bounded task consequences.

Imitation risk is evaluated under the demonstration distribution ν , but deployment histories may follow a different distribution because actions depend on π . Let μ_π^t denote the distribution of $(t, \mathcal{H}_t)$ at control step t under π . The distribution $\bar{\mu}_\pi = T^{-1} \sum_{t=0}^{T-1} \mu_\pi^t$ corresponds to choosing a control step uniformly and observing its history.

Assumption 3 (Demonstration coverage):

For the evaluated policy π , there exists a uniform constant $1 \leq C < \infty$ such that

$$\bar{\mu}_\pi(\mathcal{B}) \leq C \nu(\mathcal{B})$$

for every measurable set $\mathcal{B}$ of pairs (t, h) .

This condition excludes deployment situations absent from the reference distribution and limits how much their probabilities can be underestimated.

Assumption 4 (Action sensitivity): Let A_t^E denote the action effect with the expert policy as the reference. For a fixed action scale, there exists a uniform constant $0 \leq L < \infty$ such that

$$|A_t^E(h, a)| \leq L \|a - \pi_E(h)\|$$

for all relevant (t, h, a) .

This assumption allows action deviations to be translated into bounds on changes in task-failure probability.

Following the connection between deployment imitation error and task cost [4], Assumptions 3 and 4, together with the performance-difference identity, yield the following bound (Appendix A-B):

$$F(\pi) \leq F(\pi_E) + TL\sqrt{CR_\nu(\pi)}. \quad (9)$$

Here, C accounts for the shift from demonstration to deployment distributions, while L measures the sensitivity of task outcomes to action deviations. With other quantities fixed, a smaller $R_\nu(\pi)$ yields a tighter upper bound on task-failure probability.

This bound identifies coverage and action sensitivity as key conditions for translating imitation accuracy into task-performance guarantees. We next examine the limitations of complete-history coverage and the role of representations in risk transfer.

3) *Representation-Level Coverage and Conditional Error Discrepancy*: Assumption 3 is restrictive for complete histories and can fail after a small action deviation (Appendix A-C). Encoding histories into a task representation can merge irrelevant differences, but may also merge situations requiring different actions. This tension between distributional alignment and information loss under a noninvertible representation also appears in domain-adaptation analyses [5], [6]. The result below specializes that issue to action-risk transfer between reference and deployment distributions.

For a fixed policy, expert, and common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index, let μ_Z and ν_Z be the representation distributions induced by $\bar{\mu}_\pi$ and ν . In place of Assumption 3, suppose $\mu_Z(\mathcal{B}) \leq C_\phi \nu_Z(\mathcal{B})$ for every measurable representation event $\mathcal{B}$, with $1 \leq C_\phi < \infty$. Even at the same representation, the two distributions may contain different histories. Let $e(h) = \|\pi(h) - \pi_E(h)\|$ and define their conditional-error discrepancy as $D_\phi = \mathbb{E}_{z \sim \mu_Z} |m_\mu(z) - m_\nu(z)|$, where $m_\mu(z) = \mathbb{E}_{\bar{\mu}_\pi}[e(h)^2 \mid Z = z]$ and $m_\nu(z) = \mathbb{E}_\nu[e(h)^2 \mid Z = z]$. With square-integrable action error under both distributions and measurable versions of these conditional means,

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \leq C_\phi R_\nu(\pi) + D_\phi. \quad (10)$$

If Assumption 4 also holds, the performance-difference identity gives

$$F(\pi) \leq F(\pi_E) + TL \sqrt{C_\phi R_\nu(\pi) + D_\phi}. \quad (11)$$

Appendix A-D gives the detailed proof. The bound depends jointly on the coverage-weighted reference risk and the conditional error discrepancy; a smaller coverage factor alone need not tighten it. Appending a deterministic semantic map to retained RGB does not by itself improve coverage. For the fixed policy, expert, and encoder, D_ϕ compares conditional mean squared action errors across distributions, rather than measuring lost task information. It can vanish even when histories requiring different actions are merged, so a zero discrepancy does not establish information sufficiency. Section III-C complements risk-transfer analysis by assessing information preservation directly through task consequences; it neither bounds D_ϕ nor requires the preceding coverage conditions.

C. Semantic Inputs and Task-Supervised Representations

We now assess the preservation of decision-relevant information directly through task consequences, separating representation loss from imperfect use of the retained information. This perspective is related to information-state abstractions for partially observed control and behaviorally grounded representation learning [7], [8]. Here, information preservation is measured directly by task-failure consequences under a common continuation policy.

1) *Represented Information and Decision Capability*: Fix a reference policy π_0 and a time-history evaluation distribution μ . Define the mean failure probability associated with an action rule a as $\mathcal{L}_\mu(a) = \mathbb{E}_{(t,h) \sim \mu}[Q_t^{\pi_0}(h, a(h))]$. This quantity measures the mean failure probability after replacing the current action and following π_0 thereafter.

The complete causal history, including the time index, generates the information $\mathcal{F}_h = \sigma(\mathcal{H}_t)$. The representation $Z = \phi(\mathcal{H}_t)$ denotes the complete input determining the target command: learned features together with the target time and slot, required cached plan information, and causal scheduling records. These quantities are determined by the causal history including runtime records; the deterministic readout includes slot selection and any postprocessing used. The representation is taken before the evaluated decision readout: feedback may condition on an existing base plan, but not on its own final corrected command. It generates $\mathcal{U} = \sigma(Z) \subseteq \mathcal{F}_h$. The encodings of the base policy and the feedback branch are evaluated separately as Z . For any information set $\mathcal{F}$, let $\mathfrak{A}(\mathcal{F})$ denote all $\mathcal{F}$ -measurable action rules taking values in $\mathcal{A}$. The infimum of the mean failure probability attainable using only this information is

$$\mathcal{L}_\mu^*(\mathcal{F}) = \inf_{a \in \mathfrak{A}(\mathcal{F})} \mathcal{L}_\mu(a). \quad (12)$$

Here, $\sigma(\cdot)$ denotes the information generated by the records, and an $\mathcal{F}$ -measurable rule uses only that information. The action-rule classes are compared over the same action set, with the same reference policy followed after the current action.

An implemented action rule $\pi_Z \in \mathfrak{A}(\mathcal{U})$ is constrained both by the encoded information and by its decision mechanism. Define the gap in optimal mean failure probability due to information lost in encoding as the representation loss $\Delta_{\text{rep}}(\mathcal{U}) = \mathcal{L}_\mu^*(\mathcal{U}) - \mathcal{L}_\mu^*(\mathcal{F}_h)$, and the gap between the implemented rule and the optimum under the same representation as the decision-realization error $\epsilon_{\text{dec}}(\pi_Z) = \mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{U})$. Then

$$\mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{F}_h) = \Delta_{\text{rep}}(\mathcal{U}) + \epsilon_{\text{dec}}(\pi_Z). \quad (13)$$

This decomposition distinguishes *whether information is preserved* from *whether the preserved information is effectively used*. Both terms are nonnegative and measured in mean failure probability under reference-policy continuation. The term ϵ_{dec} includes limitations due to model capacity, residual bounds, finite data and optimization, and the mismatch between action supervision and the task objective.

2) *When Relation Errors Affect Task Decisions*: The accuracy of relation estimates must be connected to task decisions through action outcomes. We use the history-conditioned failure probability $Q_t^{\pi_0}(h, a)$ defined in (6), which already averages over unobserved physical states.

The relation variable $G \in \mathbb{R}^{d_G}$ describes geometric or semantic relations among relevant objects, and the causal context χ retains proprioceptive information and execution evidence needed for decisions. The contents of χ must be specified for the task instance, rather than set to the entire history to make the assumption automatic; unexplained effects remain in ϵ_{rel} . The representation Z admits readouts of a relation estimate $\hat{G} \in \mathbb{R}^{d_G}$ and context χ , with relation error $\delta_G = \mathbb{E}_\mu \|G - \hat{G}\|$. We use an analytical function $f(G, \chi, a)$ to approximate the conditional mean failure probability above. Here, d_G is the relation dimension.

Assumption 5 (Relation-based approximation of action outcomes): Under a fixed distribution μ , there exist a measurable function f and constants $\epsilon_{\text{rel}}, K_G \geq 0$ such that

$$\begin{aligned} \mathbb{E}_\mu \sup_{a \in \mathcal{A}} |Q_t^{\pi_0}(\mathcal{H}_t, a) - f(G, \chi, a)| &\leq \epsilon_{\text{rel}}, \\ |f(G, \chi, a) - f(\hat{G}, \chi, a)| &\leq K_G \|G - \hat{G}\|, \quad a \in \mathcal{A}. \end{aligned} \quad (14)$$

The second condition holds almost everywhere on the joint support of the true and estimated relations, and $\delta_G < \infty$. The conditional expectations, suprema, and action selections satisfy the regularity conditions in Appendix B-B.

Here, ϵ_{rel} is the approximation error due to task effects not captured by the relations and context, K_G is the sensitivity constant with respect to relation errors, and δ_G is the relation-estimation error. For example, contact and slip at the same geometry must be distinguished by the context or accounted for in the approximation error.

Proposition 1 (Representation loss in task outcomes): Under Assumption 5,

$$\begin{aligned} \Delta_{\text{rep}}(\mathcal{U}) &\leq 2(\epsilon_{\text{rel}} + K_G \delta_G), \\ \mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{F}_h) &\leq 2(\epsilon_{\text{rel}} + K_G \delta_G) + \epsilon_{\text{dec}}(\pi_Z). \end{aligned} \quad (15)$$

The proof is given in Appendix B. Comparisons between RGB and enhanced encoders use the same μ , reference continuation π_0 , action set and scale, relation definition and normalization, and context and execution inputs. Separately trained representations need not have nested information sets: the assumption and errors are evaluated for each, and smaller δ_G gives a tighter common bound only with shared bounds on ϵ_{rel} and K_G . Relation-readout accuracy alone does not rank their implemented policies. An idealized final-step correction in Appendix B-E illustrates the value of relations under nonzero, finite sensitivity; it is not a model of the actual contact dynamics. When occlusion leaves relations undefined, the relation bound applies only to the valid conditional distribution, while invalid cases remain part of the overall task results.

Relation-readout and pseudo-label errors require independent assessment (Appendix B-D); training fit cannot substitute for the evaluation errors used in the analysis.

Under the stated conditions, accurately preserving task-relevant relations and context in Z bounds representation loss; actual decision performance also depends on how the policy uses this information. Accurate relation estimates alone may not resolve ambiguities associated with motion or contact history; relevant evidence must also be retained in the causal context. Prior analyses of causal confusion and copycat behavior likewise show that observation history can contain useful information while also enabling temporally correlated shortcuts [9]–[11]. The next section examines when new observations and execution records provide additional decision value and how that value is realized through timely corrections.

D. History, Corrections, and Task Benefit

The benefit of additional execution evidence depends on the information already available and on whether the resulting correction is actually adopted.

1) Decision Value of New Observations and History: We take action-chunk execution as the concrete setting, comparing a base action supplied by an existing plan with a corrected action generated using additional execution information. This setting builds on action-chunk policies and recent methods for updating chunked plans from newer observations [12]–[14]. Under a fixed evaluation distribution μ , let t denote the target execution time, and call the action position at that control step an execution slot. Denote the action selected by the base policy for this slot by $B = \pi_0(h)$. Here, π_0 is the complete base execution rule, including plan updates, slot selection, and predefined fallback procedures. The reference policy retains the same base and feedback inference, reception, and scheduling procedures, but masks the residual at the execution output. The planning information $\mathcal{F}_p$ contains the observations and plan context used to generate and select this action, together with runtime records needed to trigger fallback, so that B is $\mathcal{F}_p$ -measurable. The current information $\mathcal{F}_c$ retains the planning information and adds the latest observations and proprioception available at time t . The complete causal history $\mathcal{F}_h$ additionally contains previous execution records, with $\mathcal{F}_p \subseteq \mathcal{F}_c \subseteq \mathcal{F}_h$. Runtime information includes only records and timestamps already available. The separation of current information from longer execution history is also motivated by prior findings that history can resolve partial observability but can be harmful when a policy exploits action correlations rather than task evidence [10], [11].

The reductions in optimal mean failure probability due to new observations and additional history are, respectively,

$$\begin{aligned} \Delta_{\text{new}} &= \mathcal{L}_\mu^*(\mathcal{F}_p) - \mathcal{L}_\mu^*(\mathcal{F}_c), \\ \Delta_{\text{hist}} &= \mathcal{L}_\mu^*(\mathcal{F}_c) - \mathcal{L}_\mu^*(\mathcal{F}_h). \end{aligned} \quad (16)$$

Both terms are nonnegative by information inclusion: a rule using less information remains available when more information is provided. For a final-decision distribution μ in a two-action setting, consider two equally likely modes with identical $\mathcal{F}_c$, including observations, proprioception, plan context, and runtime records. If history distinguishes the modes and they require opposite actions, with an incorrect choice increasing failure probability by $0 < c \leq 1$, then $\Delta_{\text{hist}} = c/2$ (Appendix B-F). If current information already determines an optimal action, the history value can instead be zero. Encoding a finite action–response history may discard decision-relevant information; the resulting loss is accounted for by the representation term below.

2) From Information Value to Implemented Corrections: The potential value of new information is not yet the benefit of a candidate correction. Applying the decomposition in (13) to the feedback representation also accounts for information loss and decision-realization error. Let Z_t^{fb} be a causal representation of the planning information, observations, execution history, and runtime records needed to determine the target-slot candidate command, generating $\mathcal{U} = \sigma(Z_t^{\text{fb}}) \subseteq \mathcal{F}_h$. As in Section III-C, this is the complete pre-readout representation: candidate selection and postprocessing are measurable functions of it. For a valid plan, let τ_p be the time of the observation used by that plan, τ_o the latest observation time

used by the candidate, and t the target execution time, with $\tau_p \leq \tau_o \leq t$. Encoding and observation delay may limit the decision-relevant information retained.

The representation produces a candidate command $\hat{u} = B + r \in \mathcal{A}$, where r is the residual for the target slot after postprocessing, including magnitude bounding. The decision-realization gap of the base action is $\epsilon_{\text{base}} = \mathcal{L}_\mu(B) - \mathcal{L}_\mu^*(\mathcal{F}_p)$.

Proposition 2 (Decomposition of candidate feedback benefit): Under the same μ , π_0 , and action set, define the candidate feedback benefit as the reduction in mean failure probability from applying the candidate correction. It satisfies

$$\begin{aligned} \Gamma_\mu &:= \mathcal{L}_\mu(B) - \mathcal{L}_\mu(\hat{u}) \\ &= \epsilon_{\text{base}} + \Delta_{\text{new}} + \Delta_{\text{hist}} - \Delta_{\text{rep}}(\mathcal{U}) - \epsilon_{\text{dec}}(\hat{u}). \end{aligned} \quad (17)$$

The proof is given in Appendix C. The decision-realization gap includes residual magnitude and capacity limits, learning error, and the mismatch between action supervision and task consequences. The matched-distribution, matched-slot action-error identity (56) diagnoses correction direction and magnitude, but does not replace this task-benefit criterion.

3) Temporal Alignment and Closed-Loop Task Benefit:

A candidate's decision advantage affects the trajectory only through its actual adoption. We connect the adoption rule to task performance using the performance-difference identity (8). Let r be the bounded target-slot candidate actually selected by the deterministic execution rule, or zero if none is available. A correction must arrive in time, remain unexpired, and match the valid plan and execution slot. The correction-adoption indicator $\Lambda \in \{0, 1\}$ equals 1 when these conditions hold and 0 otherwise. Without a valid correction, B is executed. If the base plan is also unavailable, B is the predefined fallback command for the same history, with $\Lambda = 0$ and the candidate residual set to $r = 0$. The executed command is $u_{\text{exec}} = B + \Lambda r$. Given history h , the candidate's reduction in failure probability relative to the base action is $g_t(h) = Q_t^{\pi_0}(h, B) - Q_t^{\pi_0}(h, \hat{u})$. Because the compared outcomes are failure probabilities, $g_t(h) \in [-1, 1]$. The effective benefit is $\Gamma_{\text{exec}, \mu} = \mathbb{E}_\mu[\Lambda g_t(h)]$; it depends on which corrections are adopted in time, not merely the overall adoption rate.

The reference and corrected controllers share the same computation-state transition law under Assumption 2, including the reception process, and differ only in the executed correction.

Proposition 3 (Effective feedback and task completion): Under Assumption 2, if the complete deployed policy π executes the above u_{exec} on its visited histories, while the reference policy executes $B = \pi_0(h)$ on the same histories, then

$$J(\pi) - J(\pi_0) = T \mathbb{E}_{\bar{\mu}_\pi}[\Lambda g_t(h)]. \quad (18)$$

Thus, the effective task benefit under the actual visitation distribution is positive if and only if task-completion probability exceeds that of the reference policy.

Furthermore, if Z_t^{fb} admits readouts of an estimate $\hat{G}_t$ of the execution-time relations G_t and context χ_t , and these quantities satisfy Assumption 5 under $\bar{\mu}_\pi$, the relation bound in Section III-C applies. Here, $\delta_G = \mathbb{E}_{\bar{\mu}_\pi} \|G_t - \hat{G}_t\|$ and all other

relation constants concern the feedback representation and execution time. Relation-estimation accuracy at the planning time τ_p cannot substitute for accuracy at the target execution time t (Appendix C-B). Let $p_{\text{miss}} = \mathbb{P}_{\bar{\mu}_\pi}(\Lambda = 0)$ denote the probability of nonadoption. Combining this with the timing bound gives

$$\begin{aligned} J(\pi) - J(\pi_0) &\geq T [\epsilon_{\text{base}} + \Delta_{\text{new}} + \Delta_{\text{hist}} \\ &\quad - 2(\epsilon_{\text{rel}} + K_G \delta_G) \\ &\quad - \epsilon_{\text{dec}}(\hat{u}) - p_{\text{miss}}]. \end{aligned} \quad (19)$$

A positive right-hand side means that information value and the room for improving the base decision exceed representation, realization, and timing losses, providing a sufficient condition for increased task-completion probability. All terms are evaluated under the same deployment distribution for a fixed base policy. If π_0 already includes semantics, this is a feedback-increment guarantee, not the full method's gain over RGB; local gains under different policy distributions cannot simply be added. If the relation conditions hold only on a valid subdomain, such as visible cases, the relation bound applies only to that conditional distribution. The overall benefit must also account for the invalid subdomain and cannot be guaranteed directly by (19).

This analytical sufficient condition cannot be certified from imitation and auxiliary training losses alone. The nonadoption penalty is conservative: rejected corrections need not be beneficial, and under the $r = 0$ convention a slot with no candidate has zero local adoption loss relative to its defined candidate, yet may still contribute to p_{miss} .

IV. Method

Section III shows that improving task performance from limited demonstrations depends on two related conditions: the policy representation must preserve task relations that affect action outcomes, and execution-time observations must be converted into timely and effective corrections.

To instantiate these conditions, we construct the three-stage framework illustrated in Fig. 1. First, offline quality-aware iterative annotation provides reliable supervision for semantic inputs and task relation labels. Second, a base policy jointly encodes RGB observations, soft semantics, and supervised relation queries to generate long-horizon action chunks. Finally, history-conditioned feedback uses observations acquired after plan generation to correct action slots that have not yet been executed.

A. Quality-Aware Iterative Semantic Supervision

Proposition 1 shows that the relation-estimation error δ_G enters the upper bound on representation loss measured in task outcomes. Equation (44) further decomposes this error into relation-readout error and relation pseudo-label error. Consequently, whether relation supervision provides a useful inductive bias depends not only on how well the query heads fit their supervision targets, but also on the reliability of the semantic masks from which those targets are derived.

As illustrated in Fig. 2, we first establish a high-quality seed set through limited manual review and then iteratively expand

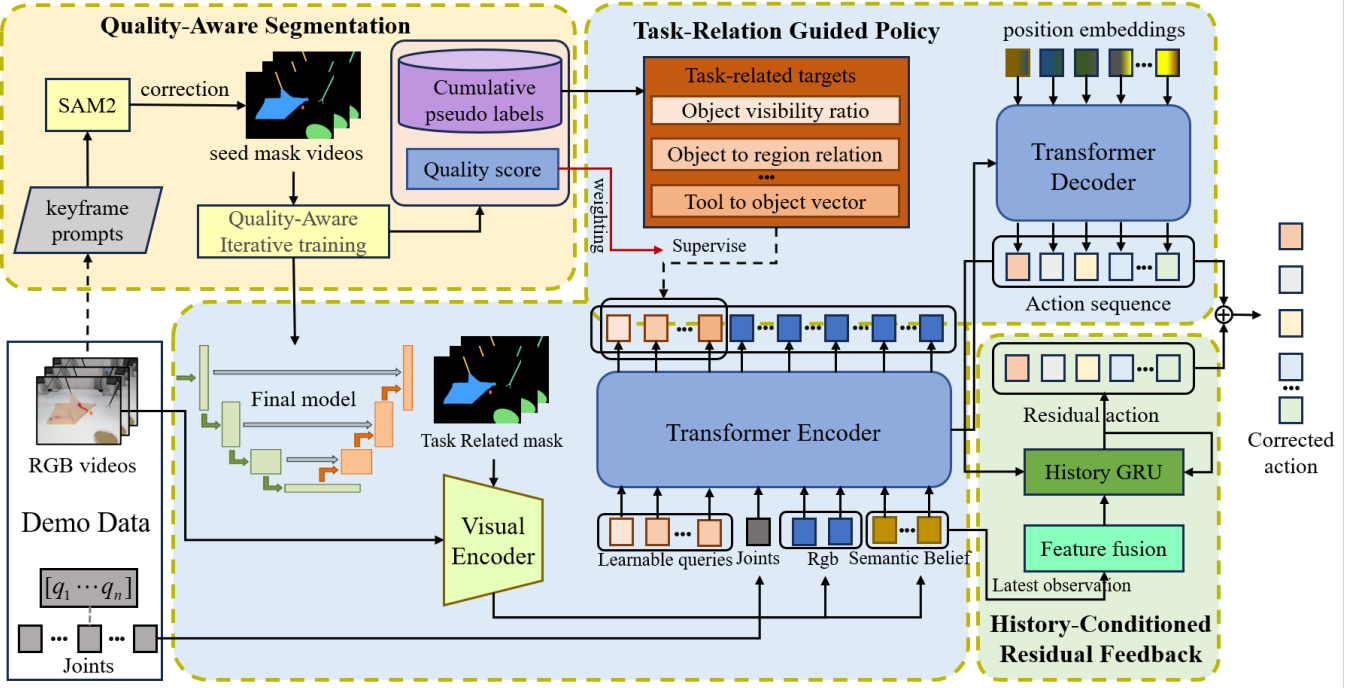


Fig. 1. Method overview. Quality-aware iterative segmentation constructs the frozen semantic model, cumulative pseudo-labels, and quality information used for weighted task-relation supervision. The base policy jointly encodes RGB observations, soft semantics, proprioception, and learnable relation queries to generate an action sequence. The history-conditioned feedback branch fuses the latest observation with the encoded policy context and predicts residual corrections for the action sequence.

semantic supervision through segmentation prediction, video-level quality screening, and the accumulation of accepted pseudo-labels across rounds.

1) Event-Guided Seed Annotation and Iterative Expansion:

For each task class, we select a small number of videos that cover the principal appearances, occlusion states, and interaction events. Keyframes are selected around substantial semantic-state changes, such as the first exposure of the target, tool-target contact, and entry of the target into the task region, rather than at fixed temporal intervals. At each keyframe, an annotator supplies a tight bounding box together with positive and negative point prompts, which SAM2 propagates both forward and backward in time [15]. Disagreement between the two propagation directions localizes unstable intervals, after which class-dependent structural constraints and manual review correct missed regions, false positives, and holes. The reviewed video masks form the initial seed set, while prompt provenance, propagation direction, and revision records are retained for traceability.

The initial seed videos are divided into training and development-validation sets at the video level, without allowing adjacent frames to cross sets, to avoid optimistic estimates caused by temporal correlation. After training the initial segmenter, each iteration produces pixelwise class probabilities on a new set of unlabeled videos $\mathcal{U}^{\text{new}}$. Mutual-exclusivity enforcement and fixed structural processing produce candidate masks, of which only those passing the quality-admission procedure below are added to the expanded set. Expanded samples accumulate across rounds, whereas the initial seed videos no longer participate in subsequent fitting and instead

serve as a fixed reference set for monitoring classwise performance, key-event performance, and catastrophic forgetting. The segmenter is reinitialized in every round, and final cross-video generalization is evaluated on separate videos that have never been used for training or model selection.

2) *Event-Preserving Video Quality Assessment*: Let $\tilde{M}_t^c$ denote the candidate mask for class c in frame t . The quality assessment retains both the raw model output and the mask after fixed structural processing: the former diagnoses model errors, whereas the latter represents the pseudo-label that is actually available for supervision. For each branch, we extract the compact state

$$\mathbf{d}_{t,c}^{\text{seg}} = \phi_{\text{seg}}(\tilde{M}_t^c), \quad (20)$$

where ϕ_{seg} summarizes mask area, centroid, bounding-box scale, connected components, and hole structure. These quantities describe class extent, location, shape, and topology, respectively, rather than serving as interchangeable criteria of correctness.

Directly constraining inter-frame differences can misclassify genuine task progress, including target exposure, contact, and pushing, as annotation errors. We therefore fit a piecewise robust expected trajectory to $\mathbf{d}_{1:T,c}^{\text{seg}}$ over the entire video. Persistent level changes form new state segments, whereas isolated or short-lived deviations provide anomaly evidence. In parallel, forward-backward-consistent optical flow establishes valid pixel correspondences between adjacent frames. Contour consistency is computed after motion-compensating the previous mask into the current frame; regions with insufficient flow evidence are marked unavailable rather than automatically as-

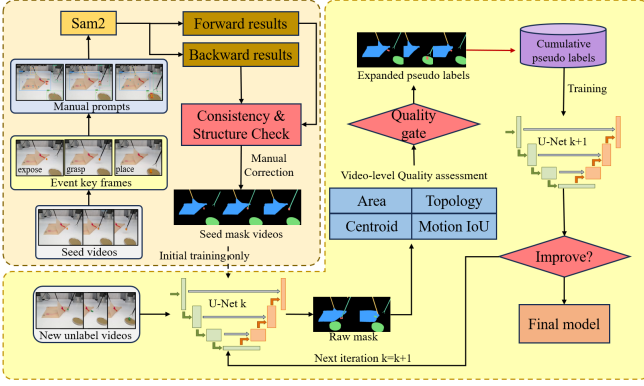


Fig. 2. Quality-aware iterative semantic annotation. Event-keyframe prompts and bidirectional SAM2 propagation are reconciled through consistency checks and manual correction to produce seed masks, which initialize only the first segmenter. Subsequent rounds predict candidate masks on new videos and apply video-level area, centroid, topology, and motion-consistency diagnostics before admitting pseudo-labels to a cumulative pool. Each round trains a reinitialized segmenter on the expanded pool, and iteration stops when the fixed validation criterion no longer improves.

signed low quality. This procedure yields four complementary evidence groups,

$$\mathbf{v}_{t,c}^{\text{seg}} = (v_{t,c}^{\text{area}}, v_{t,c}^{\text{space}}, v_{t,c}^{\text{topo}}, v_{t,c}^{\text{motion}}), \quad (21)$$

which quantify area, spatial-shape, topological, and motion inconsistency relative to the normal trajectory estimated within the video.

A class is confirmed as a hard anomaly only when multiple mutually independent evidence groups simultaneously exceed robust class- and video-specific tolerances. The detected interval is expanded over a local temporal neighborhood to prevent leakage at anomaly boundaries. If structural processing changes too many foreground pixels relative to the raw output, the frame is also excluded from the next segmentation-training round. Let $\eta_{t,c}^{\text{seg}} \in \{0, 1\}$ indicate that class c at time t has not been identified as a hard anomaly. The processed masks of a frame enter the expanded set only when the relevant classes pass the checks and the result does not depend excessively on structural processing. The continuous quality score is used only for ranking and diagnosis; it is neither interpreted as the probability of mask correctness nor supplied directly to the deployed policy.

This assessment detects deviations from the video’s own motion and structural regularities, but cannot rule out temporally stable systematic segmentation errors. The iterative model is therefore also monitored on a fixed manually reviewed reference set, and iteration stops when newly added pseudo-labels no longer improve independent validation performance. Class mappings and segmenters are defined separately for different views, and the final segmenters are frozen before policy training. SAM2 propagation and whole-video quality modeling may use subsequent frames offline, whereas online segmentation and policy inference use only observations available at the current time.

B. Task-Relation-Guided Visual-Semantic Representation

Soft semantic maps represent object classes and their spatial locations, while relation queries further constrain the representation to preserve object relations that affect decision making. Semantics therefore enters action learning through two complementary pathways: soft semantic maps provide object-level spatial structure, and supervised queries extract relations associated with task outcomes. Both pathways are combined with RGB observations, allowing the policy to retain pose, texture, and contact cues not covered by the semantic classes. Semantics obtained from the same image by a frozen segmenter provides inductive guidance without introducing additional physical observations. Its purpose is to help the joint representation preserve information relevant to control outcomes [7], [8]. Proposition 1 bounds representation loss in terms of relation-readout error under the stated conditions. Whether the proposed design improves relation readout and its use in decision making is determined experimentally.

1) *Joint RGB and Soft-Semantic Encoding*: For view v , the segmenter predicts class probabilities $p_t^v(c, \xi)$ from image I_t^v and forms a three-channel soft semantic map using fixed color vectors e_c^{sem} :

$$S_t^v(\xi) = \sum_{c \in \mathcal{C}_v} p_t^v(c, \xi) e_c^{\text{sem}}. \quad (22)$$

Here, $\mathcal{C}_v$ is the class set for view v , ξ is a pixel location, and $e_c^{\text{sem}} \in [0, 1]^3$ is the fixed color encoding of class c . An explicit class correspondence is maintained for each view. The soft color map is a compressed representation of the probabilities; it does not use argmax hardening and is not equivalent to the full multiclass probability tensor.

RGB observations, soft semantic maps, and proprioception are encoded and jointly supplied to the action-chunk policy. We instantiate the base policy with ACT [12], although the semantic representation does not depend on its particular generative structure. Multiple views remain separate input streams and are combined in the policy representation space rather than by directly averaging pixel coordinates across views.

2) *Quality-Gated Relation Query Supervision*: For a given task, we preselect K_{rel} relation groups $G^{(j)} \in \mathbb{R}^{d_j}$, such as visibility, relative-position, or region-occupancy relations, together with their coordinates, scales, and validity conditions. Together, these groups form the relation vector G in Section III-C, where d_j is the dimension of group j and $\sum_j d_j = d_G$. Fixed operators ρ_j transform the offline masks $\tilde{M}_t^{1:V}$ into relation pseudo-labels $\tilde{G}_t^{(j)} = \rho_j(\tilde{M}_t^{1:V})$. These operators are used only to construct supervision and do not provide privileged object states to the deployed policy. The specific objects, number of relations, and their computation are specified by each task instance rather than fixed by the general architecture.

Each relation group is assigned a learnable query token that enters the encoder together with visual and proprioceptive tokens. An auxiliary head reads the relation prediction $\hat{G}_t^{(j)} = D_j(z_t^{(j)})$ from its output $z_t^{(j)}$, which also remains in the encoder context available to the action decoder. Relations therefore influence the action chunk through supervised hidden representations, rather than through a hand-designed rule that

directly converts a relation value into an action or phase transition.

Video-level diagnostics must be propagated according to each relation rather than compressing the entire frame into a single quality score. Let $\mathcal{C}_j$ be the semantic classes involved in relation j , and let $\text{sel}_j(\mathbf{v}_{t,c}^{\text{seg}})$ select from (21) the evidence that directly affects this relation and aggregate it into a scalar. For example, visible area is affected primarily by target-area and topological errors, target-region distance depends on the spatial locations of both members, and the target-tool vector is additionally sensitive to tool shape and endpoint localization. The anomaly severity of relation j at time t is defined as

$$v_{t,j}^G = \max_{c \in \mathcal{C}_j} \text{sel}_j(\mathbf{v}_{t,c}^{\text{seg}}). \quad (23)$$

This relation-specific propagation prevents segmentation anomalies in irrelevant classes from disabling valid supervision and prevents an error in a critical class from being hidden by averaging across classes.

Let $m_{t,j}^G \in \{0, 1\}$ indicate whether the relation pseudo-label is geometrically and semantically defined. For a distance or directional relation, $m_{t,j}^G = 0$ whenever either member is not visible. For visible area, zero is treated as valid supervision only when target absence has been reliably confirmed; an unverified empty pseudo-mask is not equated with true zero area. The quality weight $\varpi_{t,j}^G \in [0, 1]$ combines hard-anomaly admission with relation-specific anomaly severity:

$$\varpi_{t,j}^G = m_{t,j}^G \prod_{c \in \mathcal{C}_j} \eta_{t,c}^{\text{seg}} g_j(v_{t,j}^G), \quad (24)$$

$$g_j(x) = \begin{cases} 1, & x \leq \underline{v}_j, \\ \left(\frac{\bar{v}_j - x}{\bar{v}_j - \underline{v}_j} \right)^{\gamma_j}, & \underline{v}_j < x < \bar{v}_j, \\ 0, & x \geq \bar{v}_j, \end{cases}$$

Here, $0 \leq \underline{v}_j < \bar{v}_j$ define the boundaries for full and rejected supervision, respectively, and $\gamma_j > 0$ controls the decay rate in the intermediate interval. These boundaries are determined on manually reviewed calibration videos according to the error between the pseudo-relation $\tilde{G}^{(j)}$ and the reviewed relation $G^{(j)}$, rather than tuned retrospectively against final policy success. When the calibration set is insufficient to support a continuous mapping, we set $g_j \equiv 1$, retain only hard admission, and treat continuous downweighting as a separate ablation.

Training uses $\varpi_{n,j}^G$ in the auxiliary relation loss and jointly optimizes it with the base imitation objective:

$$\mathcal{L}_{\text{rel}} = \frac{1}{K_{\text{rel}}} \sum_{j=1}^{K_{\text{rel}}} \frac{\sum_n \sum_{k=1}^{d_j} \varpi_{n,j}^G \ell_{\beta_G}(\hat{G}_{njk}^{(j)} - \tilde{G}_{njk}^{(j)})}{d_j \max(1, \sum_n \varpi_{n,j}^G)}, \quad (25)$$

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{IL}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}.$$

Here, n indexes samples in a minibatch, k indexes components of relation group j , and ℓ_{β_G} is the Smooth L1 loss with transition parameter $\beta_G > 0$. The term $\mathcal{L}_{\text{IL}}$ is the original action-chunk imitation objective of the base policy, and $\lambda_{\text{rel}} \geq 0$ weights relation supervision. Each relation group is normalized using a fixed scale. Equation (24) gives full weight to high-reliability labels, reduces the contribution of boundary

cases, and disables auxiliary supervision for clearly anomalous or undefined labels. Action supervision is retained when a relation loss is disabled, avoiding the removal of difficult but valuable action samples involving occlusion or contact. The segmenters remain frozen during this stage. The base policy retains the regularization terms and network settings of its original implementation, which are not contributions of this work.

Quality gating is used only to construct offline supervision and does not mask semantic inputs during deployment. It is distinct from the execution-validity gate Λ_t in Section IV-C. Because the selected relations need not explain every decision, both the base policy and the subsequent feedback module retain the original visual and proprioceptive information rather than treating relation values as the sole policy state.

C. History-Conditioned Residual Action Feedback

The base action chunk provides a continuous behavioral plan, while the feedback branch uses new observations and previous execution responses to correct future slots. Residuals are always defined relative to the currently valid plan, and the base policy remains frozen during both residual training and deployment.

1) *Plan-Conditioned Causal History Encoding*: At observation time τ_p , the base policy generates an action chunk together with its relation-query context. At time τ_o , the feedback branch acquires new images and generates a candidate correction for an unexecuted slot at time t , following the convention $\tau_p \leq \tau_o \leq t$. Relation queries are cached between plan updates, whereas new images are re-encoded at every feedback update.

The input record e_k^{fb} for the k th feedback update jointly encodes new image features, cached queries, proprioceptive state, a causal velocity estimate, the previously issued command, future base actions in the current plan, and relative timing information. A fixed-window GRU encodes the most recent L_{hist} records into a feedback hidden state:

$$h_i = \text{GRU}_{\omega}(e_i^{\text{fb}}, h_{i-1}), \quad i = \max(1, k - L_{\text{hist}} + 1), \dots, k. \quad (26)$$

Here, ω denotes the GRU parameters, L_{hist} is the history-window length, and h_k is the output corresponding to the last valid record. The GRU can thus interpret the current observation by associating previously issued commands with subsequently observed changes, rather than merely smoothing adjacent actions.

The cached queries describe task relations at τ_p , while the latest images and causal history provide evidence of execution changes after plan generation. Together, these inputs constitute the feedback representation Z_t^{fb} analyzed in Section III-D. The GRU is not assumed to recover the complete physical state; it is required only to preserve the action-response information needed for the current correction.

2) *Bounded Residual Prediction and Imitation Supervision*: From h_k , the feedback head predicts residuals for H_r future

slots. The candidate command for slot j combines the base action $B_{t_{k,j}}$ with a bounded residual:

$$\begin{aligned} r_{k,j} &= \beta \odot \tanh([W_r h_k + b_r]_j), \\ \hat{u}_{k,j} &= B_{t_{k,j}} + r_{k,j}, \quad j = 0, \dots, H_r - 1. \end{aligned} \quad (27)$$

Here, $t_{k,j}$ is the prescribed target execution slot, $\beta \in \mathbb{R}_{>0}^{d_u}$ contains per-action-dimension bounds, W_r and b_r are output-head parameters, and $\odot$ denotes elementwise multiplication. The quantity $r_{k,j}$ is a command residual, not an increment in semantic state, and it is not integrated cumulatively across adjacent time steps.

Residual training replays the frozen base policy along demonstration trajectories and constructs plans and target slots using the deployment timing convention. Future demonstrated actions are used only as supervision targets and do not enter the feedback input. For valid slots, we use a scale-normalized corrective action loss together with residual regularization:

$$\begin{aligned} \mathcal{L}_{\text{res}} &= \frac{1}{d_u \max(1, \sum_{n,j} m_{nj})} \sum_{n,j,c} m_{nj} \\ &\times \left[\ell_{\beta_r} \left(\frac{B_{njc} + r_{njc} - u_{njc}^E}{\sigma_c} \right) \right. \\ &\quad \left. + \lambda_r (r_{njc}/\sigma_c)^2 \right]. \end{aligned} \quad (28)$$

Here, n indexes replay samples, j indexes target slots, c indexes action dimensions, m_{nj} is the slot-validity indicator, and u_{njc}^E is the corresponding demonstrated command. The quantity $\sigma_c > 0$ is the training-set action scale, while $\beta_r > 0$ and $\lambda_r \geq 0$ are loss parameters. Offline replay does not simulate the environment response that would follow execution of a residual, so this objective learns only local corrections on the demonstration distribution.

3) *Target-Slot Alignment and Base-Plan Fallback*: Each residual is bound to the base plan and target execution slot for which it was generated and may modify only the corresponding unexecuted action. A new plan supersedes the previous plan, and late or expired residuals are discarded to prevent corrections from being applied to mismatched action chunks. The executed command is $u_t = B_t + \Lambda_t r_t$, where Λ_t indicates only plan, slot, and timing validity. When no valid residual is available, execution falls back to the base plan. The specific scheduling and timeout settings are given in Appendix D-D1.

The base action chunk thus supplies long-horizon behavioral structure, the GRU uses action–response history to generate local corrections, and slot gating ensures that each correction acts only on the future plan on which it was conditioned. Concrete relation definitions, network settings, and scheduling parameters are provided in Appendix D.

V. Experimental Validation

实验检验三个问题：完整方法是否改善任务完成，语义关系与历史反馈分别贡献什么，以及感知误差和执行延迟如何限制收益。以下给出待执行协议与结果表结构，尚无本轮实测结果。

A. Tasks and Evaluation Protocol

图 3 列出需补充的场景与执行证据。

实景素材待补
平台全景、相机/工具/目标区域位置
同一试验的拨空、重新接触、推进序列

Fig. 3. 平台与任务记录占位。最终须用可追溯的真实照片和同一试验帧替换，标明时间、对象、相机与动作方向；当前不构成实验或纠偏效果证据。

1) *Bimanual Exposure and Execution*: 评价首先在参考视角定义完整目标区域足迹 R^* ，并令 Ω 为该参考视角的整幅图像域。令 O_t^* 为目标完整投影， C_t 为遮挡布料投影。当目标完整可评估且 $|O_t^*| > 0$ 时，定义

$$p_{\text{in}}(t) = \mathbf{1}\{O_t^* \subseteq R^*\}, \quad (29)$$

$$a_C(t) = |C_t \cap \Omega|/|\Omega|, \quad (30)$$

$$p_{\text{restore}}(t) = \mathbf{1}\{a_C(t) \geq \theta_C\}. \quad (31)$$

R^* 不是当前可见 region 类别 mask：互斥分割使 object 与可见 region 不重叠，直接求交会误判。固定目标区域可由无遮挡参考帧独立标注；相机或区域移动时须重新估计。仅有局部可见的目标 mask 也不能替代 O_t^* 来证明整个物体已进入区域。

当前布料复原不会重新遮挡目标物体，复原指标是布料在画面中的面积占比，不是目标被遮挡比例，也不是默认裁剪桌面 ROI 后的占比。按这一要求，任务在规定时间内满足下式即可完成：

$$\Phi(s_{0:T}) = \mathbf{1}\{\exists t \leq T : p_{\text{in}}(t) = 1 \wedge p_{\text{restore}}(t) = 1\}. \quad (32)$$

两项必须在同一完成时刻成立：不能将不同时间分别满足、却从未共同成立的记录组合为成功。式(32)采用时限内曾共同满足的标准，不额外要求保持至评价终点；若最终协议要求终态保持或首次完成后停止，须在采集前明确并相应统一评价实现。若协议另要求连续 d_C 帧确认，则在同一窗口内检验这两个谓词，并要求窗口终点不超过 T ；未指定持续时间时不默认添加这一要求。

(待补：统一时限 T 、布料画面覆盖阈值 θ_C 与持续确认窗口 d_C ，不沿用未经核实的历史阈值。)

若最终物体与目标区域均可清晰评价，式(32)的两个谓词可以由同一参考帧判定，不必依赖假设的最终目标遮挡。执行中被机械臂临时遮挡仍可能导致评价证据不足，但不能把这种局部情况混同于任务要求布料最后重新遮住物体。暴露、分离、推动和复原是有助于解释行为的过程事件；并非所有任务都必须采用同一组过程标签，更不要在问题定义中指定阶段头。最终平台、任务与采样预算须按表 1 冻结。历史实验采用的物体和位置划分只作为候选设置，不能自动成为本轮确认事实。各方法复用示范划分、视角和匹配初始条件；语义标注、预训练和残差训练资源另行报告。已有模型的数据独立性限于附录 D-A 所核实范围。

一次完整闭环试验为任务结果记录单元。以物体、初始位置和重复编号构成配对区组，区组内随机化方法运行顺序，并保留独立训练种子。同一 checkpoint 的重复试验不是独立训练重复。主结果按 checkpoint、训练种子及物体–位置条件逐行报告成功次数/总数；Wilson 区间仅用于同一 checkpoint、同一条件下可交换且独立的重复。跨条件汇总预先固定条件权重，保留逐层计数；跨种子汇总另

TABLE I
实验协议登记；待定事项在评测前冻结。

项目	最终设置/状态
平台与任务	平台数量、动作接口和单位待定
视觉配置	相机、分辨率、标定与固定位置待定
评价规则	时限、 θ_C 、 d_C 和不可评估处理待定
物体与位置	类别、网格、训练/测试位置清单待确认
数据与重复	episode 清单、训练种子和每条件次数待定
模型与执行	checkpoint 及分割版本、重规划与延迟配置待定

固定种子权重，不将混合后的总成功数和总次数作为一个二项样本计算 Wilson 区间。跨种子的比较保留配对结构，以训练种子和条件区组为层次估计成功率差值区间；种子不足时仅作描述性报告，不用重复运行扩大有效训练样本量。试验数、停止规则、技术无效判据和统计方案须在评测前固定。

完成时间仅在成功试验中计算，同时报告失败与超时比例，不据删去失败后的均值宣称整体更快。算法超时、计划失效和有害修正均保留为失败；外部设备故障另记原因并按预定规则处理。视觉结果由独立人工审核或冻结评价器结合人工复核得到，记录分歧与不可判断比例。不可评价轨迹不静默删除；除预先定义的域内结果外，另报告将不确定结果计为失败的保守统计。

B. Task Performance and Data Budget

以 RGB/语义关系表示与无反馈/历史反馈构成交叉比较 (表 II)。同一表示下共用冻结基础 checkpoint，不同表示复用数据划分与种子。额外加入匹配调度的输出屏蔽组：保留反馈计算、接收和缓存，仅在发送动作时屏蔽残差。该组对应理论参考策略；直接停用反馈计算的组是系统基线，其差值还包含调度影响。输出屏蔽组尚待实现与核验，不能用已有无残差日志替代。报告物体与初始位置分层结果，区分示范包含与未包含的位置，不据此宣称未见类别泛化。交叉比较分别衡量表示与反馈贡献；若声称协同作用，另报告成功概率尺度上的交互差值及区间。

固定预算实验只能支持该规模下的有效性。若保留样本效率主张，须采用多个嵌套、按任务条件分层的示范预算，复用子集和训练种子。30/60/90 条仅为待确认候选，不是已完成设置。若分割器始终使用全量数据，应明确是固定外部语义资源下的机器人示范预算比较。表 III 给出预算与机制实验的必填结果字段。

C. Representation and History Ablations

表示对照依次为 RGB、RGB 加语义、再加等量无监督查询、再加关系监督查询。最后两组结构和查询数相同，以区分查询容量与监督的作用。在独立人工审核帧上，分别报告分割误差、伪标签关系误差及模型关系读出误差；按关系类型、目标大小与可见性分层，未定义关系报告有效率，不填零误差。基础查询在计划时刻 τ_p 的精度，与反馈表示对目标槽位 t 的读出精度分别评价。所有主要读出使用部署零潜变量，探针拟合与评估按 episode 分开；未来槽位图像仅作离线真值，不进入反馈输入。

历史比较固定最新观测、本体、基础动作、基础特征、槽位和训练目标，比较当前信息 MLP 与有限窗口 GRU，并

增加参数量接近的当前信息模型。如当前信息含过去观测估计的速度，需单独披露。该对照匹配 A2C2 的主要信息接口，但未经原版复现不以 A2C2 结果命名。随后分解过去视觉与执行记录的贡献；置乱历史或屏蔽查询仅作依赖性诊断，与重新训练的消融结合，避免将分布外干预直接当成因果证据。更频繁重规划和 ACT 时间集成作为执行对照，报告实际计算预算与观测年龄。

D. Mechanism and Runtime Diagnostics

离线动作比较使用同一回放分布、动作尺度、执行槽位与采样权重，同时计算 MAE/MSE、限幅饱和率和式(56)的对齐收益及修正能量。自主轨迹缺少对应专家动作时，不使用原示范的同索引动作充当真值。接触建立、拨空、滑脱和恢复事件预先定义并独立标注，用于区分关系估计错误、当前信息歧义与无效调整。这些统计解释机制，不直接估计理想历史价值或所有候选动作的后果函数。

逐槽保存观测、提交、完成和执行时间、计划 ID、基础动作、候选、采用标记及丢弃原因；报告端到端延迟分位数、更新间隔、历史窗口跨度和关键事件附近的采用率。候选动作与实际采用后动作分别诊断，避免以总体采用率替代有效反馈收益。调度为单共享工作线程、基础更新优先，目标更新频率不作为实际频率报告。缺少候选或时间日志的旧试验标记不可追溯，不补造数值。失败案例同时包括未纠正偏差与有害修正，并关联原始视频。

若需直接估计单步动作后果，仅在可复现状态及必要历史的环境中执行分支，随后沿同一参考策略续行；否则以配对闭环结果为主要证据。受控延迟和遮挡先在离线或仿真验证，实机干预另行确认安全条件。自然失败分析不得称为受控扰动实验。本章验证具体设计是否实现理论所刻画的机制，不以有限实验确认全局覆盖、稳定性或关系充分性。

VI. Discussion and Conclusion

A. Discussion

本文讨论的核心是有限机器人示范下的任务完成表现，语义引导关系表示与历史条件反馈分别针对表示不足和执行信息不足。两条路径应依据交叉对照独立评价；完整方法的提升不能替代对各环节的验证，单一数据规模的结果也不能替代样本效率曲线。离线语义资源与额外残差训练成本属于方法成本，须与性能一并报告。当前三组 front 关系是给定操作任务的实例；新任务需要重新确定对象、关系和有效性约定，并报告相应标注与训练成本。跨硬件结果不等于跨任务通用性，无多任务证据时结论限定于已验证任务。语义和几何监督提供任务相关归纳偏置，但不能补全未覆盖的接触动力学，也不能恢复观测中无法识别的任意隐藏状态。质量分数可能无法发现稳定的系统误差，历史模型也可能依赖错误相关性。当前查询监督不等于显式完成判定，监督残差亦不具备强化学习的策略改进保证。分析中的覆盖、关系对任务后果的近似充分性和后果映射正则性是适用条件，不是网络结构自动保证的性质。当前方法没有学习任务后果函数；有限动作监督能否实现理论上的信息价值，仍取决于实际决策学习与闭环验证。

式 (9) 以总体模仿风险 $R_v(\pi)$ 为输入，给出覆盖与敏感性假设下的任务失败概率上界，并非从有限示范上的训练损失直接得到的保证。若要以经验误差替代总体风险，还需控制学得策略在相同动作目标、尺度和损失下的泛化差

TABLE II

逐 checkpoint、逐条件的闭环结果空表。所有语义关系增强组仍保留 RGB 输入。每个平台/任务/物体/位置层分别实例化；seed 标识关联基础及残差训练记录。跨层总体另按预定权重汇总并报告分层区间，不混用逐层 Wilson 区间。时间条件于成功。

基础表示	执行模式	seed / checkpoint	成功/总数	成功率及 95% 区间	成功时间 (s)	超时/总数
RGB	无反馈计算	待定	待测	待测	待测	待测
RGB	历史反馈	待定	待测	待测	待测	待测
语义关系	无反馈计算	待定	待测	待测	待测	待测
语义关系	当前信息 MLP	待定	待测	待测	待测	待测
语义关系	历史 GRU	待定	待测	待测	待测	待测
语义关系	GRU 计算/输出屏蔽	待定	待测	待测	待测	待测

TABLE III

预算与机制诊断空表；组别和重复预算需在运行前确认。

对照	必填指标	状态
多示范预算	各预算/种子成功数、训练资源	待测
四组表示	成功率、关系误差/有效率	待测
当前/历史	成功率、同槽动作风险	待测
过去视觉/指令	分事件纠偏表现、输入范围	待测
重规划/时间集成	成功率、观测年龄、算力	待测
屏蔽/采用修正	成功率差值、调度分位数	待测

距，并考虑轨迹内样本相关性以及利用数据进行模型选择的影响。动作块训练损失与最终执行动作的平方误差也不一定相同，不能未经对应关系分析就直接代入该界。本文不建立从有限样本训练误差到闭环任务成功率的端到端统计保证。

数据多样性、按完整 episode 划分数据以及保留不参与模型选择的测试集，可用于降低采样局限与数据泄漏风险，并检验未见轨迹上的泛化表现；但随机划分 episode 本身不保证统计独立，这些措施也不等同于形式化泛化界。实验中应分别报告未见轨迹上的动作误差与闭环任务成功率，区分同分布泛化和新位置、物体或环境带来的分布变化，并说明分割器与策略各自的数据使用范围，避免将上游模型已使用的数据视为整个方法完全未见的测试数据。

（待补：根据实际数据划分、模型选择协议及上游分割训练记录核实上述评价范围，再结合实验结果讨论泛化差距。）

（待补讨论：依据最终失败案例说明方法在哪些条件下失效，跨平台和任务迁移的代价，以及自动标注的人工开销。）

B. Conclusion

本文围绕两个互补缺口构建研究框架：有限动作监督下任务关系可能未被充分保留，而生成动作块时的信息可能不足以支持执行中的动态修正。语义与几何监督用于构建任务关系表示，计划条件历史残差用于利用新增执行信息。条件性分析区分任务相关信息损失、历史的潜在决策价值、实际决策误差与目标槽位时效；这些依据解释可能的改善路径，不构成完整系统的无条件性能保证。

（待补正文：基于最终验证结果，简洁总结问题、方法、获得支持的结论及剩余局限；暂不声明统计显著提升或通用收敛。）

Appendix A

Task Performance and Imitation Risk

A. Performance-Difference Identity

We prove (8) by summing expected changes in the reference value. The evaluation-domain restriction in Section III-A and

Assumption 1 give the terminal value

$$V_T^{\pi_0}(\mathcal{H}_T) = 1 - \mathcal{V}(I_{0:T}^{1:V}) = 1 - \Psi. \quad (33)$$

For $t < T$, taking action $\pi(\mathcal{H}_t)$ and then following π_0 gives the failure probability $V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})$ conditional on the next history. Averaging over next histories under Assumption 2 gives

$$Q_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) | \mathcal{H}_t]. \quad (34)$$

Here, π determines the next-history distribution, while $V_{t+1}^{\pi_0}$ accounts for subsequent execution under π_0 .

Combining (7) and (34) and taking expectations gives

$$\begin{aligned} & \mathbb{E}_\pi[A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))] \\ &= \mathbb{E}_\pi[\mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) | \mathcal{H}_t]] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)]. \end{aligned} \quad (35a)$$

By the tower property, averaging the conditional next-step value over current histories gives its unconditional expectation:

$$\begin{aligned} & \mathbb{E}_\pi[\mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) | \mathcal{H}_t]] \\ &= \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})]. \end{aligned} \quad (35b)$$

Substituting (35b) into (35a) and summing over time leaves only the terminal and initial terms:

$$\begin{aligned} & \mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) \\ &= \sum_{t=0}^{T-1} \left\{ \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)] \right\} \\ &= \mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] - \mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)]. \end{aligned} \quad (35c)$$

At time T , no action remains. By (33), the terminal term is the failure probability under π :

$$\mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] = \mathbb{E}_\pi[1 - \Psi] = F(\pi). \quad (35d)$$

The policies share the initial-history distribution, so the initial term equals the failure probability of following π_0 throughout:

$$\begin{aligned} \mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)] &= \mathbb{E}_{\pi_0}[V_0^{\pi_0}(\mathcal{H}_0)] \\ &= F(\pi_0). \end{aligned} \quad (35e)$$

Finally, substituting (35d) and (35e) into (35c) gives

$$\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = F(\pi) - F(\pi_0). \quad (35f)$$

This establishes (8).

B. Proof of the Task-Failure Bound

We prove (9) by first bounding task consequences through action deviations and then transferring the resulting deployment error to the demonstration distribution.

Set $\pi_0 = \pi_E$ and let $e(h) = \|\pi(h) - \pi_E(h)\|$. Applying Assumption 4 in (8) bounds each step's increase in failure probability by its action deviation:

$$F(\pi) - F(\pi_E) \leq L \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} e(\mathcal{H}_t) \right]. \quad (36a)$$

Since $\bar{\mu}_\pi$ samples control steps uniformly, the time sum equals T times the deployment average:

$$\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} e(\mathcal{H}_t) \right] = T \mathbb{E}_{(t,h) \sim \bar{\mu}_\pi} e(h). \quad (36b)$$

By the Cauchy–Schwarz inequality, the mean action deviation is bounded by its root mean square:

$$\mathbb{E}_{\bar{\mu}_\pi} e(h) \leq \sqrt{\mathbb{E}_{\bar{\mu}_\pi} e(h)^2}. \quad (36c)$$

Assumption 3 also bounds expectations of nonnegative measurable functions, by simple-function approximation and monotone convergence. Applying it to $e(h)^2$ and using (3) gives

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \leq C \mathbb{E}_\nu e(h)^2 = C R_\nu(\pi). \quad (36d)$$

Combining (36a)–(36d) establishes (9).

C. Coverage Limitations for Complete Action Histories

The following example shows how Assumption 3 can fail for complete histories. Let $T \geq 2$. A deterministic expert always executes $u_0 = 0$ initially, whereas the learner always executes an admissible nonzero action $u_0 = \varepsilon$. Because $\mathcal{H}_1$ contains the executed action, the event $\mathcal{B} = \{t = 1, u_0 = \varepsilon\}$ satisfies

$$\bar{\mu}_{\pi_E}(\mathcal{B}) = 0, \quad \bar{\mu}_\pi(\mathcal{B}) = 1/T > 0. \quad (37)$$

Thus, for $\nu = \bar{\mu}_{\pi_E}$, we have $\bar{\mu}_\pi \not\ll \nu$, violating Assumption 3, regardless of how small ε is or how many expert demonstrations are collected. This counterexample even allows actions to have no effect on the next physical state; the mismatch arises from the recorded action history itself.

Choosing a broader reference distribution can change coverage, but does not simultaneously provide expert action labels or generalization guarantees on that distribution. Replacing the full available history by a compressed representation in the performance-difference argument would additionally require proving that it retains the task memory, controlled transitions, and action-chunk execution rules, and redefining the failure probability under reference-policy continuation accordingly. The representation-based risk transfer below instead retains complete histories for the performance-difference identity.

D. Action-Risk Transfer Through a Representation

This supplementary risk-transfer analysis is not a premise of the task-consequence representation bound in Section III-C. Related coverage–information tradeoffs appear in domain adaptation [5], [6]; the following result is derived under the present conditions. Fix the policy π , expert π_E , and a common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index. Let $\mu = \bar{\mu}_\pi$, and let μ_Z and ν_Z denote the induced distributions of Z . Assume the representation space is standard Borel so that the conditional expectations below admit measurable versions as functions of z , and that $e(h)^2 = \|\pi(h) - \pi_E(h)\|^2$ is integrable under both μ and ν . Define the conditional mean squared errors and their deployment-weighted discrepancy by

$$\begin{aligned} m_\mu(z) &= \mathbb{E}_\mu[e(h)^2 \mid Z = z], \\ m_\nu(z) &= \mathbb{E}_\nu[e(h)^2 \mid Z = z], \\ D_\phi &= \mathbb{E}_{z \sim \mu_Z} |m_\mu(z) - m_\nu(z)|. \end{aligned} \quad (38)$$

In place of Assumption 3, suppose $\mu_Z(\mathcal{B}) \leq C_\phi \nu_Z(\mathcal{B})$ for every measurable representation event, with $1 \leq C_\phi < \infty$. This condition ensures that reference null sets are also deployment null sets, so D_ϕ is independent of the versions chosen for the conditional means. Conditioning on Z and applying coverage to the nonnegative function m_ν give

$$\begin{aligned} \mathbb{E}_\mu e(h)^2 &= \mathbb{E}_{\mu_Z} m_\mu(Z) \\ &\leq \mathbb{E}_{\mu_Z} m_\nu(Z) + D_\phi \\ &\leq C_\phi \mathbb{E}_{\nu_Z} m_\nu(Z) + D_\phi \\ &= C_\phi R_\nu(\pi) + D_\phi. \end{aligned} \quad (39)$$

These assumptions also make D_ϕ finite and establish (10). If Assumption 4 also holds, the performance-difference identity gives $F(\pi) - F(\pi_E) \leq T L \mathbb{E}_\mu e(h)$. Applying Cauchy–Schwarz and (39) yields the main-text bound (11). Neither result requires Assumption 3 or a Markov assumption on Z ; only the task-performance bound additionally requires Assumption 4. If common measurable functions express both $\pi(h)$ and $\pi_E(h)$ solely in terms of Z under both distributions, then $e(h)^2$ is a common function of Z and $D_\phi = 0$. In general, D_ϕ measures a conditional mean-error difference for this fixed policy, not all decision ambiguity; $D_\phi = 0$ alone does not establish representation sufficiency. It differs from the task-consequence representation loss in Section III-C and involves expert actions on deployment histories that fixed demonstrations alone do not supply. The result is a conditional bound for fixed objects, not a uniform generalization guarantee for data-dependent training.

Measurable compression merges events: any complete-space coverage constant also bounds representation events by taking their preimages, but the converse need not hold. A measurable invertible transformation with measurable inverse preserves the best coverage constant in both directions. Appending a deterministic semantic map to retained RGB has the same property relative to RGB, since projection recovers the input. Coverage can improve through a lossy representation, with D_ϕ retaining the cost of conditional error differences; adding history need not reduce the coverage constant.

E. A Counterexample: Lower Action Risk but Worse Task Performance

Consider $T = 1$ with two equally likely, observable initial states, action set $[0, 0.2]$, and an expert that chooses 0 in both states. The failure probability in the first state is $\min(1, 100a^2)$, while the second state always leads to success. The final image directly reveals success or failure, satisfying visual assessability.

Policies π_1 and π_2 output $(0.1, 0)$ and $(0, 0.2)$ across the two states, respectively. Under the common initial distribution ν ,

$$\begin{aligned} R_\nu(\pi_1) &= 0.005 < 0.02 = R_\nu(\pi_2), \\ J(\pi_1) &= 0.5 < 1 = J(\pi_2). \end{aligned} \quad (40)$$

The coverage constant is $C = 1$, and the sensitivity bound holds with $L = 10$, since $\min(1, 100a^2) \leq 10a$. Thus, even when the bound in the main text holds, smaller action risk does not imply higher task success. The difference comes from where errors occur and their task consequences, not from an algebraic error in the bound.

Appendix B Task-Relevant Representation Analysis

A. Decision Classes and Conditional Action Outcomes

The information inclusion $\mathcal{U} \subseteq \mathcal{F}_h$ implies $\mathfrak{A}(\mathcal{U}) \subseteq \mathfrak{A}(\mathcal{F}_h)$, and hence $\mathcal{L}_\mu^*(\mathcal{U}) \geq \mathcal{L}_\mu^*(\mathcal{F}_h)$. Since the implemented rule satisfies $\pi_Z \in \mathfrak{A}(\mathcal{U})$, both gaps are nonnegative. Adding and subtracting $\mathcal{L}_\mu^*(\mathcal{U})$ in $\mathcal{L}_\mu(\pi_Z) - \mathcal{L}_\mu^*(\mathcal{F}_h)$ gives (13). This decomposition does not require the optimum to be attained by a network.

B. Regularity of Conditional Outcomes and Action Selection

Assumption 5 uses the following technical conditions: The history-conditioned $Q_t^{\pi_0}(h, a)$ uses a jointly measurable version under the common process. For any measurable action rule $a(h)$, its risk is defined directly by $\mathcal{L}_\mu(a) = \mathbb{E}_{(t,h) \sim \mu}[Q_t^{\pi_0}(h, a(h))]$. The suprema over actions in that assumption and the proof below are measurable. The approximation $\hat{Q}(Z, a) = f(\hat{G}, \chi, a)$ admits a measurable minimizing action selection. The relation-error moments and uniform approximation error are integrable as specified in the main text. These are premises of Proposition 1, not consequences of the network architecture.

The history recursion in Appendix A already averages over latent states under the common process. The evaluation measure μ reweights time–history pairs; it does not redefine the hidden-state conditional law or the reference continuation. The time index is retained in the history. If $\mathcal{A}$ is nonempty and compact, and $\hat{Q}(z, a)$ is measurable in z and continuous in a , standard measurable-selection results provide a measurable minimizing selection. The main text requires only the existence of such a selection, not continuity of the true $Q_t^{\pi_0}$ in the action.

C. From Relation Approximation to Decision Loss

Define the uniform error of the action-outcome approximation as $d = \sup_{a \in \mathcal{A}} |Q_t^{\pi_0}(\mathcal{H}_t, a) - \hat{Q}(Z, a)|$. The triangle inequality and Assumption 5 give

$$\mathbb{E}_\mu d \leq \epsilon_{\text{rel}} + K_G \delta_G. \quad (41)$$

Choose a $\mathcal{U}$ -measurable action minimizing the approximation, $\hat{a}(Z) \in \arg \min_{a \in \mathcal{A}} \hat{Q}(Z, a)$. For any $a_h \in \mathfrak{A}(\mathcal{F}_h)$, the following holds pointwise:

$$\begin{aligned} Q_t^{\pi_0}(\mathcal{H}_t, \hat{a}) &\leq \hat{Q}(Z, \hat{a}) + d \\ &\leq \hat{Q}(Z, a_h) + d \leq Q_t^{\pi_0}(\mathcal{H}_t, a_h) + 2d. \end{aligned} \quad (42)$$

Taking expectations and then the infimum over a_h yields

$$\mathcal{L}_\mu^*(\mathcal{U}) \leq \mathcal{L}_\mu(\hat{a}) \leq \mathcal{L}_\mu^*(\mathcal{F}_h) + 2(\epsilon_{\text{rel}} + K_G \delta_G). \quad (43)$$

This proves the first inequality in Proposition 1. Substitution into (13) gives the second. The rule $\hat{a}$ is used only to establish the attainable-risk bound; it does not assert that the implemented policy greedily optimizes this approximation.

D. Pseudo-Label and Relation-Readout Errors

For automatically generated relation pseudo-labels $\tilde{G}$, the pointwise triangle inequality and expectation under the same evaluation distribution give

$$\delta_G \leq \mathbb{E}_\mu \|\hat{G} - \tilde{G}\| + \mathbb{E}_\mu \|\tilde{G} - G\|. \quad (44)$$

The second term is the mean pseudo-label error relative to ground truth. Both terms require auditing under the evaluation distribution; training fit cannot replace this assessment.

E. An Idealized Relation-Based Correction Example

Restrict the evaluation distribution to the final control step of an idealized one-dimensional alignment task. The normalized lateral tool–target offset $G \in [-1, 1]$ is determined exactly by the full observation history, and $a \in [-1, 1]$ commands displacement in the same coordinates and units. There are no hidden contact modes, friction changes, or prior damage; χ is empty. After the action, a Bernoulli failure indicator is sampled with probability

$$Q_{T-1}^{\pi_0}(h, a) = f(G, a) = \frac{1}{2} + \frac{1}{8}(a - G)^2. \quad (45)$$

This probability lies in $[1/2, 1]$, and a terminal image reveals the outcome, satisfying visual assessability. This is a specified stochastic task model, not a physical model of the robot; no further action remains, so the reference continuation is immaterial. For a relation readout $\hat{G} \in [-1, 1]$, use the same f as the relation proxy. Then $\epsilon_{\text{rel}} = 0$ and

$$|f(G, a) - f(\hat{G}, a)| = \frac{1}{8}|G - \hat{G}||2a - G - \hat{G}| \leq \frac{1}{2}|G - \hat{G}|. \quad (46)$$

Thus the nonzero constant $K_G = 1/2$ is valid, and the full-information optimal action $a = G$ varies with the relation. If the implemented action is the proxy minimizer $a = \hat{G}$, then

$$\mathcal{L}_\mu(\hat{G}) - \mathcal{L}_\mu^*(\mathcal{F}_h) = \frac{1}{8}\mathbb{E}_\mu(\hat{G} - G)^2 \leq \frac{1}{4}\delta_G, \quad (47)$$

where the last inequality uses $|\hat{G} - G| \leq 2$. For equally likely offsets $G = -1$ and $G = 1$, the best constant action is 0 with failure probability $5/8$, whereas observing G permits failure probability $1/2$. The example exhibits strict decision value of a relation; it neither identifies proxy minimization with the network's training procedure nor establishes global semantic sufficiency. If unmodeled contact modes affect failure, the construction no longer applies without specified observable context or a nonzero approximation error.

F. Value of History with Identical Current Images

Consider equally likely modes $m \in \{-1, +1\}$. The evaluation distribution μ in this example is supported only at the final decision, not uniformly over the full horizon. A past action-response record reveals the mode, but the current image, relation snapshot, available plan, and runtime context contain no mode information. Thus, all of $\mathcal{F}_c$, including proprioception, is mode-independent, whereas $\mathcal{F}_h$ distinguishes the modes. One decision remains, with available actions -1 and $+1$. An action matching the mode has failure probability 0; the other action has failure probability c , where $0 < c \leq 1$. Any action based only on current information is incorrect in half the cases, giving optimal failure probability $c/2$. The complete history permits $a = m$, giving optimal failure probability 0. Consequently, $\Delta_{\text{hist}} = c/2 > 0$. This provides a sufficient case, not a guarantee for arbitrary execution histories. A final image revealing the outcome suffices to satisfy Assumption 1.

This example shows that visual assessability of completion, identical current geometry, and useful history for action selection can coexist. Conversely, if all additional histories admit the same optimal action, more information need not produce a strict task benefit. Neither case relies on an explicit stage classifier.

Appendix C Feedback and Execution Analysis

A. Decomposition of Candidate Benefit

For fixed μ and $Q_t^{\pi_0}$, $\mathcal{L}_\mu(B) = \mathcal{L}_\mu^*(\mathcal{F}_p) + \epsilon_{\text{base}}$, and (13) gives

$$\mathcal{L}_\mu(\hat{u}) = \mathcal{L}_\mu^*(\mathcal{F}_h) + \Delta_{\text{rep}}(\mathcal{U}) + \epsilon_{\text{dec}}(\hat{u}). \quad (48)$$

Subtracting these expressions and inserting $\mathcal{L}_\mu^*(\mathcal{F}_c)$, between the two optimal values yields (17). Replacing the base-action identity with $\mathcal{L}_\mu(u_c) = \mathcal{L}_\mu^*(\mathcal{F}_c) + \epsilon_c$ where u_c is an $\mathcal{F}_c$ -measurable current-information control and ϵ_c is its realization gap relative to the optimum under that information, gives

$$\mathcal{L}_\mu(u_c) - \mathcal{L}_\mu(\hat{u}) = \Delta_{\text{hist}} + \epsilon_c - \Delta_{\text{rep}}(\mathcal{U}) - \epsilon_{\text{dec}}(\hat{u}). \quad (49)$$

Combining this with Proposition 1, when the corresponding relation conditions apply to the feedback representation under the same distribution, also gives the candidate-benefit bound

$$\Gamma_\mu \geq \epsilon_{\text{base}} + \Delta_{\text{new}} + \Delta_{\text{hist}} - 2(\epsilon_{\text{rel}} + K_G \delta_G) - \epsilon_{\text{dec}}(\hat{u}). \quad (50)$$

This connects the information-preservation conditions in Section III-C to the information-use conditions in Section III-D, without dropping the unknown decision-realization error.

B. Representation and Time Alignment of Feedback Relations

For target execution time t , replace the objects in Proposition 1 by $Z = Z_t^{\text{fb}}$, $G = G_t$, $\hat{G} = \hat{G}_t(Z_t^{\text{fb}})$ and $\chi = \chi_t(Z_t^{\text{fb}})$. These readouts use only causal inputs available when the candidate is generated. The true relations G_t define the analytical error; they are not additional state inputs supplied to the feedback network. Relation visibility, outcome approximation, and regularity conditions must hold for these objects under the same μ . The original proof then applies, but its error quantities cannot be identified with those of the base encoder at planning time.

For example, suppose the stale base-query estimate $\hat{G}_{\tau_p}^{\text{base}}$ is used as the current relation readout. With matching dimensions and coordinate conventions, and relations defined at both times,

$$\begin{aligned} \mathbb{E}_\mu \|G_t - \hat{G}_{\tau_p}^{\text{base}}\| &\leq \mathbb{E}_\mu \|G_t - G_{\tau_p}\| \\ &\quad + \mathbb{E}_\mu \|G_{\tau_p} - \hat{G}_{\tau_p}^{\text{base}}\|. \end{aligned} \quad (51)$$

The first term is the change in relations from the planning observation to the target execution time. The second is still weighted by the sampling distribution induced by deployment target slots and is not the geometric training loss. Even with an exact old estimate, the first term may be large. Latest RGB inputs and history are intended to provide evidence of this change, but including them alone does not bound the readout error. Without time-aligned relation validation, the candidate-benefit decomposition and closed-loop identity remain applicable, but base-query error cannot be substituted directly into (19).

C. Adoption at Target Slots and Task Outcomes

Reception order, plan validity, and fallback-trigger records are included in the causal history. The last accepted candidate for the same plan and slot is selected, followed by an age check. This overlap-resolution rule does not guarantee that the selected correction has a larger task benefit. The reference rule π_0 and deployed rule share the same fallback procedure at a given history when the base plan becomes invalid. For this comparison, the reference retains feedback computation, reception, and cache updates and masks only the correction sent to the control interface. Both controllers use the same computation-state transition law, including the distribution of runtime delays conditional on the available history. Thus, base-slot selection gives the same B at the same available history; realized plans and completion times need not coincide across different closed-loop trajectories. An implementation of this comparator must keep scheduling and candidate reception independent of the output-mask flag. Skipping residual inference changes this process and requires a separate system-level comparison rather than the residual-only identity below. If the procedure terminates an episode, its outcome is fixed under the common evaluation protocol and the horizon is completed with an absorbing state; the trial is not discarded. If the actual fallback command differs from $B = \pi_0(h)$, it must be included in the execution rule and the argument rechecked; the following binary-choice identity does not apply directly.

At each time–history pair, B is executed when $\Lambda = 0$, and $\hat{u}$ is executed when $\Lambda = 1$. Thus, without assuming that $Q_t^{\pi_0}$ is linear in the action,

$$Q_t^{\pi_0}(h, B) - Q_t^{\pi_0}(h, B + \Lambda r) = \Lambda g_t(h). \quad (52)$$

Candidate selection is a deterministic function of the inputs and reception records available for that slot, with $r = 0$ if no candidate is available. Results that have not passed reception checks are excluded; accepted candidates must still pass age and plan checks at execution time. Under the shared runtime recording process, candidate selection and the adoption indicator are causal rules. Taking expectations gives the effective benefit used in the main text:

$$\Gamma_{\text{exec}, \mu} = \mathbb{E}_{\mu}[\Lambda g_t(h)]. \quad (53)$$

Since $V_t^{\pi_0}(h) = Q_t^{\pi_0}(h, B)$, (8), with the trajectory sum written as T times its deployment average, yields Proposition 3.

The bound $|g_t| \leq 1$ also yields the conservative timing bound

$$\Gamma_{\text{exec}, \mu} \geq \Gamma_{\mu} - \mathbb{P}_{\mu}(\Lambda = 0). \quad (54)$$

This bound treats each nonadoption as potentially losing the maximum benefit and is generally loose. The event-level quantity $\mathbb{E}_{\mu}[(1 - \Lambda)g_t]$ better explains the specific effect. Validity checks may reject harmful corrections, so nonadoption need not always reduce benefit.

D. Differences Between Replay and Deployment Distributions

Fix the candidate rule, reference policy, and validity rule. Let ν denote the replay evaluation distribution and $\mu = \bar{\mu}_{\pi}$ the deployment time-mixture distribution. If, on the same time–history space, $\|\mu - \nu\|_{\text{TV}} \leq \epsilon_{\text{dist}}$, then

$$J(\pi) - J(\pi_0) \geq T(\mathbb{E}_{\nu}[\Lambda g_t] - 2\epsilon_{\text{dist}}). \quad (55)$$

Here, total variation is defined by $\|\mu - \nu\|_{\text{TV}} = \sup_{\mathcal{B}} |\mu(\mathcal{B}) - \nu(\mathcal{B})|$, where $\mathcal{B}$ ranges over the measurable events of that space. Since $\Lambda g_t \in [-1, 1]$, the difference between its expectations under the two distributions is at most twice their total variation; combining this with (18) gives the result. Identical initial positions do not ensure similar execution-history distributions; the total-variation distance between complete trajectory distributions may approach 1. This bound identifies a condition needed to extrapolate from replay; it does not claim that the current data provide a nonvacuous numerical guarantee.

E. Auxiliary Diagnostics from Action Supervision

On a separate fixed replay distribution ν_{lab} with reliable expert labels, define the expert command U^E , base error $E = U^E - B$, and residual r using the same action units and target slots, and assume that the relevant second moments are finite. The change in squared action risk satisfies

$$\mathbb{E}_{\nu_{\text{lab}}} \|E\|^2 - \mathbb{E}_{\nu_{\text{lab}}} \|E - r\|^2 = 2\mathbb{E}_{\nu_{\text{lab}}} \langle E, r \rangle - \mathbb{E}_{\nu_{\text{lab}}} \|r\|^2. \quad (56)$$

Replacing r by Λr gives the corresponding identity for adopted corrections. This identity checks whether corrections move toward the demonstrated actions and whether their magnitude

is excessive, but it does not contain the counterfactual failure probabilities $Q_t^{\pi_0}$. Improved action alignment therefore cannot replace the actual task benefit required by (18), nor does it establish stability.

Appendix D

Method Details and Implementation Status

A. 数据来源与划分口径

本附录以本地保存的 Qtoken-FS 及其 SF-C-GRU 配置为可核对实例，不将其自动认定为最终实验版本。基础训练配置指向 `bettersetup-v5`，入口读取传入数据根目录的全部轨迹；配置没有列出训练、验证和测试 episode ID，上游是否独立划分仍需原始清单确认。`front` 与 `side` 分割清单分别指向 `bettersetup-v4` 和 `bettersetup-v5`，各记载 15118 个训练样本、2819 个验证样本，但两个列表均仅保留前 2000 项。其中的视频文件索引也不能直接等同于 episode ID。这些记录能证明存在分割器内部划分，不能证明它与策略使用同一划分。

残差缓存生成器按任务分层，以固定种子划分约 80% 训练和 20% 验证轨迹，并明确将验证范围限定为残差头；冻结基础策略可能已见全部轨迹。本地保存了 SF-C-GRU 的缓存指纹，但未找到该缓存的原始划分文件，故不能重建其确切 episode 名单，也不能由种子猜测名单。各阶段证据及缺失项登记在随稿的 `data_split_audit.json`；其未知名单使用空值，不能当作已经完成的统一划分。

后续全流程独立实验须先建立共同的轨迹清单，记录原始采集 ID、数据版本、各阶段用途及派生文件映射，再由提示器、分割器、基础策略和残差训练复用。若固定已有上游模型，只重新划分残差数据，则结果继续限定为残差头独立验证。当前证据既不足以断言最终测试泄漏，也不足以声称全流程未见。

B. 质量门控与已有实现边界

本文统一方法以整段视频趋势诊断生成帧级、类别级质量状态，并将其作为伪标签和关系标签的离线准入门控。该诊断使用完整视频，不能在部署时读取，也不解释为在线置信概率。最终实验须据此重新生成关系监督并重新训练基础策略。

现有迭代分割器的开发记录已实现种子退出训练、扩展数据跨轮累计和分割器重新初始化；但其内部训练–验证仍采用通过门控后的帧级随机划分。正文规定的视频级划分和独立最终测试须在定稿实验中重新执行，否则内部验证指标不能解释为跨视频泛化性能。种子视频训练过初始教师，因此后续只能作为固定参考集，不能作为独立最终测试集。

已有 Qtoken-FS checkpoint 使用的是早期启发式质量分数及连续软加权，没有读取新的整段视频质量状态，不能视为已经实现本文统一门控。该旧实现的面积项仍将空目标掩码作为零面积监督，尚不能可靠区分确认不可见、漏标和标注失败。因此，既有结果用于形成方法假设和初步对照；最终结论须以采用新门控重训后的匹配实验为准。正式式 (23)–(24) 所述的关系相关证据选择、硬异常排除与校准降噪权尚未集成到该 checkpoint；最终实现还须保存每组关系的校准集合、阈值、衰减参数及有效样本比例，并与仅硬准入的版本分别报告。概率温度校准和扰动集成不属于当前方法。

TABLE IV
Qtoken-FS 记录中的语义颜色与类别顺序 (8-bit RGB), 不是最终配置
默认值。

类别	RGB 编码	视角
background	(0, 0, 0)	front/side
occluder	(64, 160, 255)	front/side
object	(255, 105, 97)	front/side
region	(119, 221, 119)	front/side
tool	(255, 209, 102)	front/side
leftarm	(234, 146, 199)	front
rightarm	(173, 214, 101)	front

C. 关系查询的任务实例

已有查询配置采用三组 front 图像关系：目标可见面积占比、目标到目标区域的质心距离，以及目标质心相对工具端点的位置。在宽 W 、高 H_I 的 front 图像中，离线硬标签的目标、区域和工具掩码分别为 M_O, M_R, M_T ，非空掩码的像素质心为 c_O, c_R, c_T 。三个关系的具体算子为

$$\begin{aligned}\tilde{G}^{(1)} &= |M_O|/(WH_I), \\ \tilde{G}^{(2)} &= \|c_O - c_R\|/\sqrt{(W-1)^2 + (H_I-1)^2}, \\ \tilde{G}^{(3)} &= \text{clip}(\bar{c}_O - \bar{e}_{T,L}, -1, 1).\end{aligned}\quad (57)$$

其中， $\bar{c} = (c_x/(W-1), c_y/(H_I-1))$ 为归一化质心， $\bar{e}_{T,L}$ 为归一化工具端点，clip 逐分量裁剪。坐标原点位于图像左上角，横轴向右、纵轴向下；因此 $\tilde{G}^{(3)}$ 的两个分量分别为目标相对端点向右、向下的偏移。

工具端点由归一化工具像素的协方差矩阵主特征向量 a_T 确定。对工具像素 ξ ，计算投影 $a_T^\top(\xi - \bar{c}_T)$ 的极值 α_-, α_+ ，得到 $\bar{e}_- = \bar{c}_T + \alpha_- a_T$ 与 $\bar{e}_+ = \bar{c}_T + \alpha_+ a_T$ 。取横坐标较小者为 $\bar{e}_{T,L}$ ，横坐标相同时沿用实现选择 $\bar{e}_-$ 。它不自动等同于物理接触端；方向变化可能改变这一约定的适用性，须在任务实例中核验。区域质心距离也不等同于物体被完整包含，因此任务完成评价仍采用独立定义。

已有 Qtoken-FS 实现根据旧掩码质量分数和几何有效性确定监督权重。面积组的有效标记恒为 1，空目标掩码直接生成零面积标签；位置组要求对应目标和区域或工具掩码的像素质量均大于 10^{-6} 。实现以占位数值填充无效关系，但由零权重排除其损失。它尚未读取正文所述的整段视频质量门控。当前三组输出维度为 1、1、2，每个查询为 512 维，采用 $\beta_G = 0.01$ 的 Smooth L1 损失。已有实现将关系所涉及类别的质量分数最小值记为 κ_{nj} ，以有效标记 m_{nj}^G 构造组内权重 $w_{njc} = m_{nj}^G \min(\kappa_{nj}/0.95, 1)$ 。这里 κ_{nj} 为启发式质量分数，不是校准后的正确概率。分组归一化主要改变样本的相对贡献；当分母不受截断影响时，全组同倍降权会被分母抵消。

Qtoken-FS 保存的类别颜色见表 IV。实现中的 palette 为浮点数；表内 8-bit RGB 各通道除以 255 得到式(22)的 $e_c^{\text{sem}} \in [0, 1]^3$ ，再按类别概率加权，不先取 argmax 或量化为整数。通道按表 IV 中各视角可用类别的顺序排列。该实例的 front 含机械臂类别，side 不含；不能笼统称为所有视角均未分割机械臂。保存的语义温度为 1，未拟合附录中的概率校准参数。已有基础网络使用完整视野的 480×270 图像、共享 ResNet18 骨干及 ACT 投影与位置编码；双视角含四路视觉输入，单视角含两路。当前该配置不额外加入视角或模态 embedding，也不包含阶段分类或停止头。保存配置中的动作损失系数和关系损失系数均为 1；ACT 默认 KL 系数为 10，训练日志亦满足 $L_{\text{action}} = L_{\text{L1}} + 10L_{\text{KL}}$ 。

因此，该记录在式(25)中取 $\lambda_{\text{rel}} = 1$ ，而 $\mathcal{L}_{\text{IL}}$ 内部沿用上述 ACT 目标。冻结分割器在基础策略阶段的分割损失为零。这些值对应现有 Qtoken-FS 记录；最终采用不同 checkpoint 时须重新核对。

1) ACT 潜变量的训练与部署：基础训练启用 CVAE 时，由本体观测与示范动作序列构造后验并采样潜变量；其投影 token 与关系查询、图像和本体 token 进入同一编码器。关系辅助头读取该编码器的查询输出，故其训练损失也受后验潜变量条件影响。测试时潜变量置零，沿用 ACT 的约定 [12]。残差缓存适配器将冻结基础策略置于 eval() 模式，并仅向动作块预测接口提供图像与本体输入，缓存查询也来自这次部署模式前向计算。后续关系误差评价须使用相同模式，另报告训练后验条件与零潜变量条件的差异；不将训练关系损失直接作为部署关系精度，也不在测量前断言存在动作信息捷径。

D. 残差训练与执行设置

已有完整反馈变体使用双视角 RGB 与缓存关系查询。新图像完整缩放至 320×180 ，经冻结骨干编码，每视角特征池化为 $2 \times 3 \times 512$ ；仅 RGB 与额外分割描述作为不同输入变体。当前记录使用 128 维投影、单层 hidden128 的 GRU 和 8 条历史记录。目标反馈频率为 5Hz，等间隔满窗口的跨度约 1.4 秒；在线间隔受共享线程调度影响，须由实际时间戳报告。每次从零状态重算窗口，而不是跨 episode 保持无限递归记忆。当前信息 MLP 使用同一记录格式中的最新一条，保留计划、本体和即时运动上下文。

监督指令来自 leader，观测关节来自 follower。动作与残差使用数据集原生命单位，关节顺序、动作维度及单位须与部署一致。每维训练集动作标准差以 10^{-3} 为下限作为 σ_c ；残差限幅取训练集基础动作误差绝对值的 95 分位数，再裁剪到 $[0.05, 5]$ 原生单位。这些值是当前实现约定，不是跨平台通用安全阈值。残差训练采用 $\beta_r = 0.1$ 、 $\lambda_r = 10^{-3}$ ，AdamW 学习率 10^{-4} 、batch128、最多 50 轮，验证 8 轮不改善早停，以验证集尺度归一化 MAE 选择 checkpoint。

SF-C-GRU 保存配置的控制频率为 30Hz，基础动作块为 60 步，基础更新目标间隔为 30 步；反馈目标间隔为 6 步，预留 3 步延迟，预测 9 步修正。在第 6 帧获取的观察对应第 9-17 帧，第 12 帧观察对应第 15-23 帧，重叠 3 帧，使用首槽的端到端预算为 100ms。该记录中的观测年龄、残差年龄、历史断流阈值及视角时间差上限分别为 0.15 秒、0.4 秒、0.3 秒和 0.04 秒。历史断流阈值允许正常的 0.2 秒目标间隔，若实际资源等待超过 0.3 秒，历史仍会被清空。离线缓存按固定步长及预设延迟构造样本，没有重演在线基础任务优先与共享线程忙碌造成的反馈等待。因此，5Hz 与 1.4 秒仅为目标设置及其名义窗口跨度，不是实测执行频率。

完整时效分析应记录每个候选的观察时间、提交与完成时间、计划编号、目标槽位、接收状态及最终执行状态；现有日志字段不能覆盖的量须另补采集，不回填推测值。延迟统计包含采集、预处理、编码、排队、GRU 和指令发布，不以网络计算耗时代替端到端时效。无残差时复用有效基础计划；计划过期时当前桥接代码抛出异常并退出策略。异常之后的硬件处理仍需单独验证，不将抛出异常视为安全保证。

1) 时间戳与目标槽位：采用控制频率 f_{ctrl} 和运行起点 s_{origin} ，第 k 次反馈的图像参考时间为 $a_k = \min_v(s_k^v -$

$s_{\text{origin}})$, 其中 s_k^v 为视角 v 的采集完成时间戳。预留延迟 d_{fb} 以控制步长, 候选槽位按当前实现映射为

$$t_{k,j} = \text{round}[f_{\text{ctrl}}(a_k + d_{\text{fb}}/f_{\text{ctrl}})] + j. \quad (58)$$

其中 $j = 0, \dots, H_r - 1$, round 采用代码中的 Python 最近整数规则。在与控制网格对齐的离线帧上, $n_k = f_{\text{ctrl}}a_k$ 为整数, 该式化为 $t_{k,j} = n_k + d_{\text{fb}} + j$ 。在线时间戳不一定落在网格上, 应保留式(58)的舍入顺序。

执行循环以相对墙钟 s 确定全局槽位 $\text{round}(f_{\text{ctrl}}s)$; 基础动作块以自己的观察时间 a_p 为起点, 读取相对索引 $\text{round}(f_{\text{ctrl}}(s - a_p))$ 。这两次舍入与候选槽位映射分别记录, 不直接相减已舍入时间戳替代相对索引。实际控制循环可能跳过槽位, 未执行槽位的候选不顺延使用。基础预测与反馈共用单工作线程和一个待完成任务, 到期基础更新优先; 反馈在空闲且观测有效时才提交。

E. 实现边界与复现记录

已有代码提供掩码时序诊断、软颜色语义输入、front 关系查询监督、固定窗口 GRU/MLP 残差及计划槽位校验。新增的人工校准划分、温度拟合、扰动关系诊断与其完整审核报告尚未实现, 原有模型也不能视为已经使用这些机制。最终实验须记录分割与策略权重、颜色及类别表、数据划分、关系定义、残差输入变体和全部执行配置, 并区分已有结果与采用新质量流程重新训练、重新验证的结果。

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